Patrick Mockridge

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PROFILE AND METHOD

Protocol engineer. Process safety engineer, 2006–2015; chartered 2013; independent protocol and verification work since 2016.

The method: specify the failure mode before the architecture; prove what can be proved; contain what cannot; record provenance with every result. DarkWow is containment; Rholo is continuity; R-Cash is composition; R-Gen is claims; R-Share is capacity; R-Sure is risk; Azoth is provenance; Mudra is verification in a familiar form.

SELECTED WORK

DarkWow

Designed for private organisations on a public chain, without an administrator or a public voter roll

2026–; redesign April–October 2026 · Research prototype (mainnet TBD); genesis-set contracts; one developer with AI; mechanized circuit gate; HAZOP tabletop

Rholo

A Rust node for the ρ-calculus: native object capabilities and machine-checked laws

2026; port August–October 2026 (about two months) · Pre-release; v0.1.0 testnet tag; 91 laws (131 entries), 122 proved including 36 node-tied; 14 crates, 0 unsafe; community testnet live

R-Cash

A Wise-shaped multi-currency account and payments rail on Rholo

2026; October–November (v0.5) · Devnet-only prototype; 13/13 devnet suites; Lean 4 and five TLA+ models

R-Gen

Rholang contracts and Lean 4 proofs for engineering claims and K-Assets

2026; November · Devnet-only; 7 terms; 10/10 suites; 131 Lean theorems

R-Share

On-chain booking and rental primitives, settled through R-Cash

2026; November · Devnet-only; 13 terms; 16/16 suites; Lean allocator

R-Sure

A Lloyd’s-style insurance market on Rholo, settled through R-Cash

2026; November · Devnet-only; 17 contracts; 37 suites; 261 Lean theorems

Azoth

A process calculation library where every number carries its provenance

2026 · v0.1.x; 97 upstream issues filed (80 closed); no-op Gibbs minimizer and wrong-unit vapour-pressure curve; not for design work

Mudra

Prove a committed calculation trace is unchanged without revealing the spreadsheet

2026 · Research build; ~6 KB proof; insurance brokering and fiscal metering case studies

Records: patrickmockridge.github.io/darkwow/ · /rholo/ · /r-cash/ · /r-gen/ · /r-share/ · /r-sure/ · /azoth/ · /mudra/

CHRONOLOGY

PERIOD	WORK
2026	Delivery: Rholo port (August–October), DarkWow redesign (April–October), R-Cash (October–November), R-Gen/R-Share/R-Sure (November), Azoth, Mudra.
2022–2026	Independent research and writing: network analysis, influence/secrecy simulation; RChain legacy work: Gorki/BigSur fork, RGov/Rhobot proposal with Jim Whitescarver (2024); UNISS/Ingenesist simulations (2026).
2019–2022	Created the Romeo–Juliet Framework; continued RChain work; independent research through COVID; property renovation; crypto investments.
2016–2019	Treehouse AI and Robotics nanodegrees; early ERC-1155 at Enjin (2016); named advisor, Enjin Coin ICO (2017); CoEngineers investor (2017–2019).
2006–2015	Chemical engineer, process safety; UK and North Africa. Chartered 2013. Bitcoin since 2011; London Ethereum community, 2014.
2002–2006	MEng Chemical Engineering, University of Edinburgh.